

Challenge: BlackEnergy Lab

Platform: SOC Analyst Tier 2

Category: Endpoint Forensics

Tactics: Privilege Escalation, Stealth

Tools: Volatility

VIEWS LAB: [link](#)

Kịch bản: Develop practical skills in Windows memory forensics using Volatility by detecting malware indicators, analyzing suspicious processes, and identifying code injection and unauthorized DLLs in a compromised system.Q1

Q1

Which volatility profile would be best for this machine?

```
MINGW64/c/Users/votha/Downloads/99-BlackEnergy/temp_extract_dir
leads, linux.library_list.LibraryList, linux.lsmode.Lsmode, linux.lsof.Lsof, linux.malfind.Malfind, lin
ux.malware.check_afinfo.Check_afinfo, linux.malware.check_creds.Check_creds, linux.malware.check_idt
.Check_idt, linux.malware.check_modules.Check_modules, linux.malware.check_syscall.Check_syscall, li
linux.malware.hidden_modules.Hidden_modules, linux.malware.keyboard_notifiers.Keyboard_notifiers, linu
x.malware.malfind.Malfind, linux.malware.modxview.Modxview, linux.malware.netfilter.Netfilter, linux
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Variable Value
Kernel Base 0x804d7000
DTB 0x39000
Symbols file:///C:/Users/votha/AppData/Local/Programs/Python/Python312/Lib/site-packages/volatility3/symbols/windows/ntoskrnl.
pdb/47A5AC97343A4A7ABF14FFD9E9933772-2.json.xz
Is64Bit False
IsPAE False
layer_name 0 WindowsIntel
memory_layer 1 FileLayer
KdDebuggerDataBlock 0x8054cde0
NTBuildLab 2600.xpsp.080413-2111
CSDVersion 3
KdVersionBlock 0x8054cdb8
Major/Minor 15.2600
MachineType 332
KeNumberProcessors 1
SystemTime 2023-02-13 18:29:11+00:00
NtSystemRoot C:\WINDOWS
NtProductType NtProductWinNt
NtMajorVersion 5
NtMinorVersion 1
PE MajorOperatingSystemVersion 5
PE MinorOperatingSystemVersion 1
PE Machine 332
PE TimeDateStamp Sun Apr 13 19:27:39 2008
vvThais@ThaiHocHanh MINGW64 ~/Downloads/99-BlackEnergy/temp_extract_dir (main)
$
```

- **NTBuildLab:** 2600.xpsp.080413-2111 → Đây là phiên bản **Windows XP** (Build 2600).
- **CSDVersion:** 3 → Service Pack 3 (**SP3**).
- **Is64Bit:** False → Kiến trúc **32-bit** (x86).

Q2

How many processes were running when the image was acquired?

```
MINGW64~/Users/votha/Downloads/99-BlackEnergy/temp_extract_dir
vThais@ThaiHochanh MINGW64 ~/Downloads/99-BlackEnergy/temp_extract_dir (main)
$ vol -f "CYBERDEF-567078-20230213-171333.raw" windows.plist
Volatility 3 Framework 2.28.0
Progress: 100.00
PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output
0 System 0x89c037f8 55 245 N/A False N/A N/A Disabled
368 4 smss.exe 0x89965020 3 19 N/A False 2023-02-14 04:54:15.000000 UTC N/A Disabled
392 368 csrss.exe 0x89a98da0 11 321 0 False 2023-02-14 04:54:15.000000 UTC N/A Disabled
316 368 winlogon.exe 0x89a88da0 18 508 0 False 2023-02-14 04:54:15.000000 UTC N/A Disabled
360 616 services.exe 0x89938998 15 240 0 False 2023-02-14 04:54:15.000000 UTC N/A Disabled
372 616 lsass.exe 0x89aa0020 21 335 0 False 2023-02-14 04:54:15.000000 UTC N/A Disabled
332 660 VBoxService.exe 0x89aaa3d8 9 115 0 False 2023-02-14 04:54:15.000000 UTC N/A Disabled
380 660 svchost.exe 0x89aab590 21 295 0 False 2023-02-13 17:54:16.000000 UTC N/A Disabled
368 660 svchost.exe 0x89a9f6f8 10 244 0 False 2023-02-13 17:54:17.000000 UTC N/A Disabled
1060 660 svchost.exe 0x89730da0 51 1072 0 False 2023-02-13 17:54:17.000000 UTC N/A Disabled
1108 660 svchost.exe 0x897289a8 5 78 0 False 2023-02-13 17:54:17.000000 UTC N/A Disabled
1156 660 svchost.exe 0x899adda0 13 192 0 False 2023-02-13 17:54:17.000000 UTC N/A Disabled
1484 1440 explorer.exe 0x89733938 14 489 0 False 2023-02-13 17:54:18.000000 UTC N/A Disabled
1608 660 spoolsv.exe 0x897075d0 10 106 0 False 2023-02-13 17:54:18.000000 UTC N/A Disabled
380 1060 wscntfy.exe 0x89694388 1 28 0 False 2023-02-13 17:54:30.000000 UTC N/A Disabled
340 660 alg.exe 0x8969d2a0 5 102 0 False 2023-02-13 17:54:30.000000 UTC N/A Disabled
376 1484 VBoxTray.exe 0x89982da0 13 125 0 False 2023-02-13 17:54:30.000000 UTC N/A Disabled
336 1484 msmsgs.exe 0x8994a020 2 157 0 False 2023-02-13 17:54:30.000000 UTC N/A Disabled
1880 1484 taskmgr.exe 0x89a0b2f0 0 - 0 False 2023-02-13 18:25:15.000000 UTC 2023-02-13 18:26:21.000000 UTC
964 Disabled rootkit.exe 0x899dd740 0 - 0 False 2023-02-13 18:25:26.000000 UTC 2023-02-13 18:25:26.000000 UTC
960 Disabled cmd.exe 0x89a18da0 0 - 0 False 2023-02-13 18:25:26.000000 UTC 2023-02-13 18:25:26.000000 UTC
528 Disabled notepad.exe 0x896c5020 0 - 0 False 2023-02-13 18:26:55.000000 UTC 2023-02-13 18:27:46.000000 UTC
1432 Disabled notepad.exe 0x89a0d180 0 - 0 False 2023-02-13 18:28:25.000000 UTC 2023-02-13 18:28:40.000000 UTC
1444 Disabled notepad.exe 0x899e6da0 0 - 0 False 2023-02-13 18:28:42.000000 UTC 2023-02-13 18:28:47.000000 UTC
276 1484 DumpIt.exe 0x89a0fda0 1 25 0 False 2023-02-13 18:29:08.000000 UTC N/A Disabled
vThais@ThaiHochanh MINGW64 ~/Downloads/99-BlackEnergy/temp_extract_dir (main)
$ |
```

Có tổng cộng là 25 tiến trình trừ đi 6 tiến trình đã bị Disabled thì còn có 19 tiến trình.

Q3

What is the process ID of cmd.exe?

```

vvThais@ThaiHocAnh MINGW64 ~/Downloads/99-BlackEnergy/temp_extract_dir (main)
$ vol -f "CYBERDEF-567078-20230213-171333.raw" windows.pslist
Volatility 3 Framework 2.28.0
Progress: 100.00

```

PID	PPID	ImageFileName	PDB	scanning	finished	offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
4	0	System	0x89c037f8	55	245	N/A	False	N/A	N/A	N/A	Disabled		
368	4	smss.exe	0x89965020	3	19	N/A	False	N/A	N/A	N/A	2023-02-14 04:54:15.000000 UTC	N/A	Disabled
592	368	csrss.exe	0x89a98da0	11	321	0	False	N/A	N/A	N/A	2023-02-14 04:54:15.000000 UTC	N/A	Disabled
616	368	winlogon.exe	0x89a88da0	18	508	0	False	N/A	N/A	N/A	2023-02-14 04:54:15.000000 UTC	N/A	Disabled
660	616	services.exe	0x89938998	15	240	0	False	N/A	N/A	N/A	2023-02-14 04:54:15.000000 UTC	N/A	Disabled
672	616	lsass.exe	0x89aa0020	21	335	0	False	N/A	N/A	N/A	2023-02-14 04:54:15.000000 UTC	N/A	Disabled
832	660	VBoxService.exe	0x89aaa3d8	9	115	0	False	N/A	N/A	N/A	2023-02-14 04:54:15.000000 UTC	N/A	Disabled
880	660	svchost.exe	0x89aab590	21	295	0	False	N/A	N/A	N/A	2023-02-13 17:54:16.000000 UTC	N/A	Disabled
968	660	svchost.exe	0x89a9f6f8	10	244	0	False	N/A	N/A	N/A	2023-02-13 17:54:17.000000 UTC	N/A	Disabled
1060	660	svchost.exe	0x89730da0	51	1072	0	False	N/A	N/A	N/A	2023-02-13 17:54:17.000000 UTC	N/A	Disabled
1108	660	svchost.exe	0x897289a8	5	78	0	False	N/A	N/A	N/A	2023-02-13 17:54:17.000000 UTC	N/A	Disabled
1156	660	svchost.exe	0x899adda0	13	192	0	False	N/A	N/A	N/A	2023-02-13 17:54:17.000000 UTC	N/A	Disabled
1484	1440	explorer.exe	0x89733938	14	489	0	False	N/A	N/A	N/A	2023-02-13 17:54:18.000000 UTC	N/A	Disabled
1608	660	spoolsv.exe	0x897075d0	10	106	0	False	N/A	N/A	N/A	2023-02-13 17:54:18.000000 UTC	N/A	Disabled
480	1060	wscntfy.exe	0x89694388	1	28	0	False	N/A	N/A	N/A	2023-02-13 17:54:30.000000 UTC	N/A	Disabled
540	660	alg.exe	0x8969d2a0	5	102	0	False	N/A	N/A	N/A	2023-02-13 17:54:30.000000 UTC	N/A	Disabled
376	1484	VBoxTray.exe	0x89982da0	13	125	0	False	N/A	N/A	N/A	2023-02-13 17:54:30.000000 UTC	N/A	Disabled
636	1484	msmsgs.exe	0x8994a020	2	157	0	False	N/A	N/A	N/A	2023-02-13 17:54:30.000000 UTC	N/A	Disabled
1880	1484	taskmgr.exe	0x89a0b2f0	0	-	0	False	N/A	N/A	N/A	2023-02-13 18:25:15.000000 UTC	2023-02-13 18:26:21.000000 UTC	
964	1484	rootkit.exe	0x899dd740	0	-	0	False	N/A	N/A	N/A	2023-02-13 18:25:26.000000 UTC	2023-02-13 18:25:26.000000 UTC	
1960	964	cmd.exe	0x89a18da0	0	-	0	False	N/A	N/A	N/A	2023-02-13 18:25:26.000000 UTC	2023-02-13 18:25:26.000000 UTC	Dis
528	1484	notepad.exe	0x896c5020	0	-	0	False	N/A	N/A	N/A	2023-02-13 18:26:55.000000 UTC	2023-02-13 18:27:46.000000 UTC	
1432	1484	notepad.exe	0x89a0d180	0	-	0	False	N/A	N/A	N/A	2023-02-13 18:28:25.000000 UTC	2023-02-13 18:28:40.000000 UTC	
1444	1484	notepad.exe	0x899e6da0	0	-	0	False	N/A	N/A	N/A	2023-02-13 18:28:42.000000 UTC	2023-02-13 18:28:47.000000 UTC	
276	1484	DumpIt.exe	0x89a0fda0	1	25	0	False	N/A	N/A	N/A	2023-02-13 18:29:08.000000 UTC	N/A	Disabled

```

vvThais@ThaiHocAnh MINGW64 ~/Downloads/99-BlackEnergy/temp_extract_dir (main)
$

```

Dựa vào bảng trên có thể xác định được cmd.exe có số PID là 1960

Q4

What is the name of the most suspicious process?

```

MINGW64/C:/Users/yotha/Downloads/99-BlackEnergy/temp_extract_dir

```

32\lsass.exe	C:\WINDOWS\system32\lsass.exe	C:\WINDOWS\system32\lsass.exe	
*** 660 616	services.exe	C:\WINDOWS\system32\services.exe	
32\services.exe	C:\WINDOWS\system32\services.exe	C:\WINDOWS\system32\services.exe	
*** 832	VBoxService.exe	C:\WINDOWS\system32\VBoxService.exe	
S\system32\VBoxService.exe	C:\WINDOWS\system32\VBoxService.exe	C:\WINDOWS\system32\VBoxService.exe	
*** 1060	svchost.exe	C:\WINDOWS\system32\svchost.exe	
S\system32\svchost.exe	C:\WINDOWS\system32\svchost.exe	C:\WINDOWS\system32\svchost.exe	
*** 480	wscntfy.exe	C:\WINDOWS\system32\wscntfy.exe	
S\system32\wscntfy.exe	C:\WINDOWS\system32\wscntfy.exe	C:\WINDOWS\system32\wscntfy.exe	
*** 1156	svchost.exe	C:\WINDOWS\system32\svchost.exe	
S\system32\svchost.exe	C:\WINDOWS\system32\svchost.exe	C:\WINDOWS\system32\svchost.exe	
*** 968	svchost.exe	C:\WINDOWS\system32\svchost.exe	
S\system32\svchost.exe	C:\WINDOWS\system32\svchost.exe	C:\WINDOWS\system32\svchost.exe	
*** 1608	spoolsv.exe	C:\WINDOWS\system32\spoolsv.exe	
S\system32\spoolsv.exe	C:\WINDOWS\system32\spoolsv.exe	C:\WINDOWS\system32\spoolsv.exe	
*** 880	svchost.exe	C:\WINDOWS\system32\svchost.exe	
S\system32\svchost.exe	C:\WINDOWS\system32\svchost.exe	C:\WINDOWS\system32\svchost.exe	
*** 1108	svchost.exe	C:\WINDOWS\system32\svchost.exe	
S\system32\svchost.exe	C:\WINDOWS\system32\svchost.exe	C:\WINDOWS\system32\svchost.exe	
*** 540	alg.exe	C:\WINDOWS\system32\alg.exe	
32\alg.exe	C:\WINDOWS\system32\alg.exe	C:\WINDOWS\system32\alg.exe	
1484	explorer.exe	C:\WINDOWS\system32\explorer.exe	
*** 964	rootkit.exe	C:\WINDOWS\system32\rootkit.exe	
*** 1960	cmd.exe	C:\WINDOWS\system32\cmd.exe	
1\WINDOWS\system32\cmd.exe	C:\WINDOWS\system32\cmd.exe	C:\WINDOWS\system32\cmd.exe	
* 1432	notepad.exe	C:\WINDOWS\system32\notepad.exe	
skVolume1\WINDOWS\system32\notepad.exe	C:\WINDOWS\system32\notepad.exe	C:\WINDOWS\system32\notepad.exe	
* 1444	notepad.exe	C:\WINDOWS\system32\notepad.exe	
skVolume1\WINDOWS\system32\notepad.exe	C:\WINDOWS\system32\notepad.exe	C:\WINDOWS\system32\notepad.exe	
* 528	notepad.exe	C:\WINDOWS\system32\notepad.exe	
skVolume1\WINDOWS\system32\notepad.exe	C:\WINDOWS\system32\notepad.exe	C:\WINDOWS\system32\notepad.exe	
* 1880	taskmgr.exe	C:\WINDOWS\system32\taskmgr.exe	
skVolume1\WINDOWS\system32\taskmgr.exe	C:\WINDOWS\system32\taskmgr.exe	C:\WINDOWS\system32\taskmgr.exe	
* 276	DumpIt.exe	C:\WINDOWS\system32\DumpIt.exe	
erDefenders\Desktop\DumpIt-moonsoils\DumpIt.exe	C:\WINDOWS\system32\DumpIt.exe	C:\WINDOWS\system32\DumpIt.exe	
* 376	VBoxTray.exe	C:\WINDOWS\system32\VBoxTray.exe	
32\VBoxTray.exe	C:\WINDOWS\system32\VBoxTray.exe	C:\WINDOWS\system32\VBoxTray.exe	
* 636	msmsgs.exe	C:\WINDOWS\system32\msmsgs.exe	
Messenger\msmsgs.exe	C:\WINDOWS\system32\msmsgs.exe	C:\WINDOWS\system32\msmsgs.exe	

```

vvThais@ThaiHocAnh MINGW64 ~/Downloads/99-BlackEnergy/temp_extract_dir (main)
$

```

Nó không nằm ở system mà là tiến trình con của CyberDefenders\desktop

Q5

Which process shows the highest likelihood of code injection?

```
MINGW64/c/Users/votha/Downloads/99-BlackEnergy/temp_extract_dir
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 2a 00 2a 00 01 00 00 00 00 00 00 00 ....*.
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
616 winlogon.exe 0x7f6f0000 0x7f7effff Vad PAGE_EXECUTE_READ 0 0 Disabled
N/A
c8 00 00 00 84 01 00 00 ff ee ff ee 08 70 00 00 .....p..
08 00 00 00 00 fe 00 00 00 10 00 00 20 00 00 .....
00 02 00 00 00 20 00 00 8d 01 00 00 ff ef fd 7f .....
03 00 08 06 00 00 00 00 00 00 00 00 00 00 00 00 .....
0 08 00 00 00 00 fe 00 00 00 10 00 00 20 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
880 svchost.exe 0x980000 0x988fff VadS PAGE_EXECUTE_READWRITE 9 1 Disabled
MZ header
4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 MZ.....
b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 .....@.....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 f8 00 00 00 .....
0 b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 .....
00 00 00 00 00 f8 00 00 00 00 00 00 00 00 00 00 .....
636 msmsgs.exe 0x490000 0x78ffff Vad PAGE_EXECUTE_READ 0 0 Disabled
N/A
c1 00 00 00 00 01 00 00 ff ee ff ee 09 00 00 00 .....
09 00 00 00 00 fe 00 00 00 10 00 00 20 00 00 .....
00 02 00 00 00 20 00 00 1b 1f 00 00 ff ef fd 7f .....
00 00 08 06 00 00 00 00 00 00 00 00 00 00 00 00 .....
0 09 00 00 00 00 fe 00 00 00 10 00 00 20 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
vvThais@ThaiHocHanh MINGW64 ~/Downloads/99-BlackEnergy/temp_extract_dir (main)
$ AC
vvThais@ThaiHocHanh MINGW64 ~/Downloads/99-BlackEnergy/temp_extract_dir (main)
$ |
```

Tất cả các tiến trình còn lại có thể không thể xác định được tên của nhà cung cấp tiến trình svchost.exe lại xác định được: MZ header

Một giá trị 2 byte (0x4D5A) xác định tệp là tệp thực thi theo định dạng MZ. Chúng ta có thể chứng minh điều đó không?

Q6

There is an odd file referenced in the recent process. Provide the full path of that file.

```
mingw64/c:/Users/votha/Downloads/99-BlackEnergy/temp_extract_dir
880 svchost.exe 0xe1b11760 0x2e8 Port 0x1f0001 -
880 svchost.exe 0x8968c770 0x2ec Thread 0x1f03ff Ttd 380 Pid 880
880 svchost.exe 0xe19917a0 0x2f0 Port 0x1f0001 -
880 svchost.exe 0xe1b116a8 0x2f4 Port 0x1f0001 -
880 svchost.exe 0xe1b1a020 0x2f8 Port 0x1f0001 -
880 svchost.exe 0xe1a15d48 0x2fc Token 0xf01ff -
880 svchost.exe 0x899dfca0 0x300 Event 0x100003 -
880 svchost.exe 0x899a24b0 0x304 File 0x12013f \Device\Termd
880 svchost.exe 0x899a3df8 0x308 Event 0x1f0003 -
880 svchost.exe 0x89695ab8 0x30c Thread 0x1f03ff Ttd 236 Pid 880
880 svchost.exe 0xe1adf218 0x310 Port 0x1f0001 -
880 svchost.exe 0xe1aee2e0 0x314 Port 0x1f0001 -
880 svchost.exe 0xe1b30318 0x318 Port 0x1f0001 -
880 svchost.exe 0xe1ba5770 0x31c Port 0x1f0001 -
880 svchost.exe 0x896c4240 0x320 Event 0x1f0003 -
880 svchost.exe 0x8968c020 0x324 Thread 0x1f03ff Ttd 1836 Pid 880
880 svchost.exe 0x89a71f58 0x32c Event 0x100003 -
880 svchost.exe 0x896862b8 0x330 Mutant 0x1f0001 {3D5A1694-CC2c-4ee7-A3D5-A879A9E3A623}
880 svchost.exe 0x89a0da50 0x334 Thread 0x1f03ff Ttd 1704 Pid 880
880 svchost.exe 0x89b9d840 0x338 Mutant 0x1f0001 -
880 svchost.exe 0x89a0f0c0 0x340 File 0x12013f \Device\HarddiskVolume1\WINDOWS\system32\drivers\str.sys
880 svchost.exe 0xe1115570 0x344 Key 0xf003f MACHINE\SOFTWARE\CLASSES
880 svchost.exe 0xe1115570 0x348 Key 0xf003f MACHINE\SOFTWARE\CLASSES
880 svchost.exe 0x899892d0 0x352 Key 0xf003f MACHINE\SOFTWARE\CLASSES
880 svchost.exe 0xe1bb4cf0 0x350 Key 0xf003f MACHINE\SOFTWARE\MICROSOFT\COM3
880 svchost.exe 0x89986dc8 0x354 Event 0x1f0003 -
880 svchost.exe 0xe1b0e968 0x358 Key 0x10 USER -
880 svchost.exe 0x89a7b930 0x35c Event 0x1f0003 -
880 svchost.exe 0xe11b8468 0x360 Key 0xf003f MACHINE\SOFTWARE\CLASSES
880 svchost.exe 0x89ba1628 0x364 Event 0x1f0003 -
880 svchost.exe 0xe16b73c8 0x368 Key 0x10 USER -
880 svchost.exe 0x899e1f18 0x36c Event 0x1f0003 -
880 svchost.exe 0xe2086a88 0x370 Key 0xf003f MACHINE\SOFTWARE\MICROSOFT\COM3
880 svchost.exe 0x89a492b0 0x374 Event 0x1f0003 -
880 svchost.exe 0xe2086af0 0x378 Key 0xf003f MACHINE\SOFTWARE\MICROSOFT\COM3
880 svchost.exe 0x89b9d880 0x37c Event 0x1f0003 -
880 svchost.exe 0xe1b4a3c8 0x380 Key 0xf003f MACHINE\SOFTWARE\CLASSES\CLSID
880 svchost.exe 0x8969fc30 0x384 Event 0x1f0003 -
880 svchost.exe 0xe1b4a360 0x388 Key 0xf003f MACHINE\SOFTWARE\CLASSES
880 svchost.exe 0x89c0a498 0x38c Event 0x1f0003 -
880 svchost.exe 0xe15b7360 0x390 Key 0xf003f MACHINE\SOFTWARE\MICROSOFT\COM3
880 svchost.exe 0x89bd3db0 0x394 Event 0x1f0003 -
880 svchost.exe 0xe19b8140 0x398 Key 0x10 USER -
880 svchost.exe 0x89b961e0 0x39c Event 0x1f0003 -
880 svchost.exe 0xe16a0fb8 0x3a0 Key 0xf003f MACHINE\SOFTWARE\MICROSOFT\COM3
880 svchost.exe 0x89b9d118 0x3a4 Event 0x1f0003 -
880 svchost.exe 0xe16a0f50 0x3a8 Key 0xf003f MACHINE\SOFTWARE\MICROSOFT\COM3
880 svchost.exe 0x89aac2c0 0x3ac Event 0x1f0003 -
```

Tôi phát hiện ra điều này thật kỳ lạ. Chúng ta cũng có thể tìm thấy điều này trong đầu ra chuỗi của tiến trình dumped.\Device\HarddiskVolume1\WINDOWS\system32\drivers\str.sys

➔ KEY: C:\WINDOWS\system32\drivers\str.sys

Q7

What is the name of the injected DLL file loaded from the recent process?

```
mingw64/c:/Users/votha/Downloads/99-BlackEnergy/temp_extract_dir
OWS\system32\taskmgr.exe - - -
* 276 1484 DumpIt.exe 0x89a0fda0 1 25 0 False 2023-02-13 18:29:08.000000 UTC N/A - "C:\Documents and Settings\CyberDefenders\De
sktop\dumpit-moonsols\DumpIt.exe" C:\Documents and Settings\CyberDefenders\Desktop\dumpit-moonsols\DumpIt.exe
* 376 1484 VBoxTray.exe 0x899892d0 13 13 0 False 2023-02-13 17:54:30.000000 UTC N/A \Device\HarddiskVolume1\WINDOWS\system32\VBoxTray.ex
e "C:\WINDOWS\system32\VBoxTray.exe" C:\WINDOWS\system32\VBoxTray.exe
* 636 1484 msmsgs.exe 0x8994a020 2 157 0 False 2023-02-13 17:54:30.000000 UTC N/A \Device\HarddiskVolume1\Program Files\Messenger\msms
gs.exe "C:\Program Files\Messenger\msmsgs.exe" /background C:\Program Files\Messenger\msmsgs.exe

v\This@ThaHochiang mingw64 - Downloads/99-BlackEnergy/temp_extract_dir (main)
$ vol -f "C:\CYBERDEF-567078-20230213-171333.raw" windows.ldrmodules --pid 880
Volatility 3 Framework 2.28.0
C:\Users\votha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\deprecation.py:28: FutureWarning: This API (volatility3.plugins.windows.malwa
re.ldrmodules.LdrModules.run) will be removed in the first release after 2026-06-07. This plugin has been renamed, please call volatility3.plugins.windows.malware.ldrmodule
s.LdrModules rather than volatility3.plugins.windows.ldrmodules.LdrModules.
warnings.warn(

Pid Process Base InLoad InInit InMem MappedPath
880 svchost.exe 0x6f880000 True True True \WINDOWS\AppPatch\AcGenral.dll
880 svchost.exe 0x1000000 True False True \WINDOWS\system32\svchost.exe
880 svchost.exe 0x6700000 True True True \WINDOWS\system32\xpss2res.dll
880 svchost.exe 0x8900000 False False False N/A
880 svchost.exe 0x9a00000 False False False \WINDOWS\system32\msxml3r.dll
880 svchost.exe 0x6c00000 True True True \WINDOWS\system32\shimeng.dll
880 svchost.exe 0x5b860000 True True True \WINDOWS\system32\netapi32.dll
880 svchost.exe 0x5d090000 True True True \WINDOWS\system32\comctl32.dll
880 svchost.exe 0x68000000 True True True \WINDOWS\system32\rsaenh.dll
880 svchost.exe 0x7c900000 True True True \WINDOWS\system32\ntdll.dll
880 svchost.exe 0x7d000000 True True True \WINDOWS\system32\advapi32.dll
880 svchost.exe 0x76f60000 True True True \WINDOWS\system32\wldap32.dll
880 svchost.exe 0x76b40000 True True True \WINDOWS\system32\winmm.dll
880 svchost.exe 0x769c0000 True True True \WINDOWS\system32\userenv.dll
880 svchost.exe 0x71b70000 True True True \WINDOWS\system32\samlib.dll
880 svchost.exe 0x71ab0000 True True True \WINDOWS\system32\ws2_32.dll
880 svchost.exe 0x71aa0000 True True True \WINDOWS\system32\ws2help.dll
880 svchost.exe 0x71a50000 True True True \WINDOWS\system32\mswsock.dll
880 svchost.exe 0x71ad0000 True True True \WINDOWS\system32\wsock32.dll
880 svchost.exe 0x760f0000 True True True \WINDOWS\system32\termsrv.dll
880 svchost.exe 0x74f70000 True True True \WINDOWS\system32\icaapi.dll
880 svchost.exe 0x74980000 True True True \WINDOWS\system32\msxml3.dll
880 svchost.exe 0x722b0000 True True True \WINDOWS\system32\sensapi.dll
880 svchost.exe 0x75110000 True True True \WINDOWS\system32\mstlsapi.dll
880 svchost.exe 0x76a80000 True True True \WINDOWS\system32\rpcss.dll
880 svchost.exe 0x76b00000 True True True \WINDOWS\system32\atl.dll
880 svchost.exe 0x76c90000 True True True \WINDOWS\system32\imagehlp.dll
880 svchost.exe 0x76c30000 True True True \WINDOWS\system32\wintrust.dll
880 svchost.exe 0x76bc0000 True True True \WINDOWS\system32\regapi.dll
880 svchost.exe 0x76e10000 True True True \WINDOWS\system32\adsldpc.dll
```

Tiêm DLL phản chiếu: Tiêm DLL phản xạ là một kỹ thuật cho phép kẻ tấn công chèn DLL vào tiến trình nạn nhân từ bộ nhớ thay vì từ đĩa. Chúng ta có thể dùng plugin và chỉ định PID của tiến trình đó để lấy nó. Plugin ldrmodules liệt kê tất cả các DLL đã được tải vào không gian bộ nhớ của tiến trình được chỉ định, cùng với địa chỉ cơ sở, kích thước và đường dẫn trên hệ thống tệp. Thông tin này có thể hữu ích trong việc xác định bất kỳ DLL độc hại nào có thể đã được chèn vào bộ nhớ của tiến trình hoặc xác định các mô-đun gây ra hành vi bất thường.

Q8

What is the base address of the injected DLL?

```

MINGW64/c/Users/votha/Downloads/99-BlackEnergy/temp_extract_dir
e 376 1484 VBoxTray.exe 0x89982da0 13 125 0 False 2023-02-13 17:54:30.000000 UTC N/A \Device\HarddiskVolume1\WINDOWS\system32\VBoxTray.exe
e 636 1484 msmsgs.exe 0x8994a020 2 157 0 False 2023-02-13 17:54:30.000000 UTC N/A \Device\HarddiskVolume1\Program Files\Messenger\msmsgs.exe

vvThais@ThaiHochanh MINGW64 ~/Downloads/99-BlackEnergy/temp_extract_dir (main)
$ vol -f ".\CYBERDEF-567078-20230213-171333.raw" windows.ldrmodules --pid 880
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C:\Users\votha\AppData\Local\Programs\Python\Python312\Lib\site-packages\volatility3\framework\deprecation.py:28: FutureWarning: This API (volatility3.plugins.windows.malware.ldrmodules.LdrModules.run) will be removed in the first release after 2026-06-07. This plugin has been renamed, please call volatility3.plugins.windows.malware.ldrmodule
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880 svchost.exe 0x9a0000 False False False \WINDOWS\system32\msxml3r.dll
880 svchost.exe 0x5ad70000 True True True \WINDOWS\system32\uxtheme.dll
880 svchost.exe 0x5cb70000 True True True \WINDOWS\system32\shimgsp.dll
880 svchost.exe 0x5b860000 True True True \WINDOWS\system32\netapi32.dll
880 svchost.exe 0x5d090000 True True True \WINDOWS\system32\comctl32.dll
880 svchost.exe 0x68000000 True True True \WINDOWS\system32\rsaenh.dll
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880 svchost.exe 0x75110000 True True True \WINDOWS\system32\mstlsapi.dll
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880 svchost.exe 0x76c30000 True True True \WINDOWS\system32\wintrust.dll
880 svchost.exe 0x76bc0000 True True True \WINDOWS\system32\regapi.dll
880 svchost.exe 0x76e10000 True True True \WINDOWS\system32\adslpcc.dll
880 svchost.exe 0x76ee0000 True True True \WINDOWS\system32\rasapi32.dll
880 svchost.exe 0x76e90000 True True True \WINDOWS\system32\rasman.dll
880 svchost.exe 0x76e80000 True True True \WINDOWS\system32\ruttlis.dll

```

Chúng ta có thể lấy địa chỉ căn cứ.0x980000